

NETFLIX — REQUIREMENTS BREAKDOWN

Functional & Non-Functional Requirements

Functional Requirements
Register, Login & Authentication
Browse Content
Search Content
Multi-Resolution Video Streaming
Personalized Recommendations
Continue Watching Across Devices
Multiple Profile Management
Offline Downloads
Ratings & Watchlist

Non-Functional Requirements
High Availability
Scalability
Reliability
Low Latency
Fault Tolerance
Performance
Security
Global Content Delivery

3. Capacity Estimation

Before designing the architecture, we estimate the expected traffic, storage, and bandwidth requirements. These estimations help us choose the right CDN, storage, database, cache, and scaling strategy.

Assumptions

Parameter	Assumption
Registered Users	300 Million
Daily Active Users	100 Million
Daily Video Views	1 Billion
Average Video Size	500 MB
Read : Write Ratio	1000 : 1

3.1 Storage Estimation

Assume around 100,000 new videos are uploaded every day, and the average video size is 500 MB.

Step	Value
Daily Storage = 100,000 x 500 MB	50 TB / day
Monthly Storage = 30 x 50 TB	1.5 PB
Estimated Storage (excluding replicas and backups)	1.5 PB / month

3.2 Bandwidth Estimation

Assume the platform streams approximately 1 billion video views per day, with an average streamed size of 500 MB.

Step	Value
Daily Data Transfer = 1 Billion x 500 MB	500 PB / day
Bandwidth = 500 PB / 86,400 seconds	approx. 46 Tb/s
Estimated Bandwidth	~46 Tb/s

Note: Most of this bandwidth is handled through geographically distributed CDNs rather than the origin servers.

3.3 Server Estimation

Assume a single streaming server can handle approximately 50,000 concurrent streaming sessions.

Step	Value
Number of Streaming Servers = 100 Million / 50,000	2,000 Servers
Estimated Streaming Servers (excluding CDN edge servers)	2,000 Servers

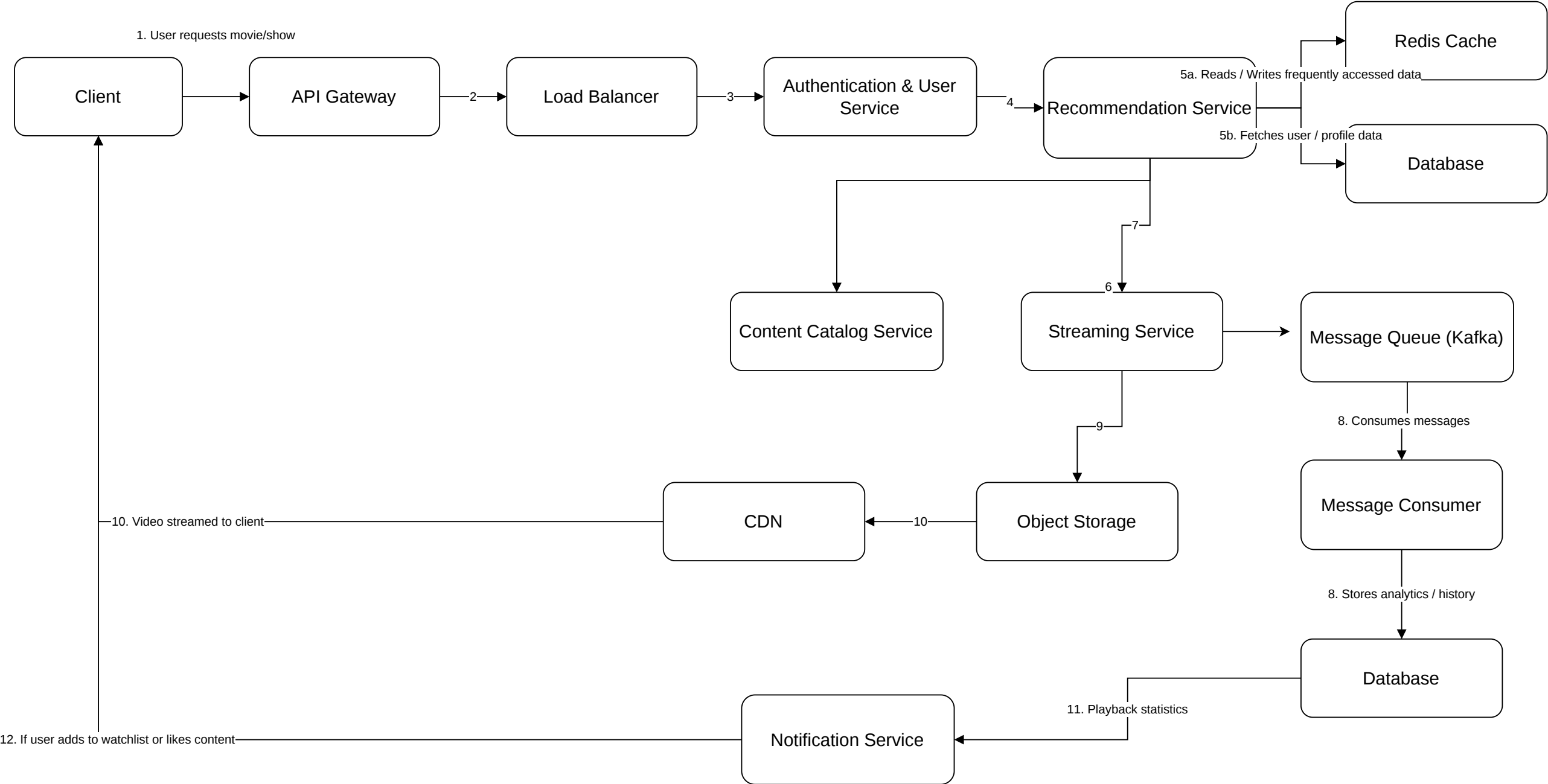
3.4 Requests Per Second (RPS) Estimation

To estimate the traffic handled by the platform, calculate the average number of streaming requests processed every second. Assume Netflix serves 1 billion video play requests per day.

Step	Value
RPS = 1 Billion / 86,400	approx. 11,574 req/s (~11.5K RPS)
Estimated Traffic	~11.5K RPS (higher at peak)

Note: This is an average estimation. During peak hours (evenings, weekends, or new content releases), actual RPS can be several times higher, so the system should be designed to handle sudden traffic spikes efficiently.

Netflix System - Request Flow Diagram



Netflix System Architecture

